

BENNINGTON MORSE, VT, USA

WMO: 726166

Lat: 42.891N

Lon: 73.247W

Elev: 826

StdP: 14.26

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 54781

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.7	(c) 1.0	(d) -14.4	(e) 2.6	(f) -2.3	(g) -9.2	(h) 3.4	(i) 3.7	(j) 24.1	(k) 42.4	(l) 19.8	(m) 36.7	(n) 2.9	(o) 280	(p) 0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	86.6	72.1	83.9	70.5	81.6	69.3	74.3	83.2	72.8	81.3	71.2	79.1	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.5	119.9	79.9	70.0	113.8	78.0	68.4	107.6	76.3	38.3	83.3	36.9	81.2	35.5	79.4	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.4	13.5	-12.1	91.1	5.4	1.8	-16.0	92.4	-19.2	93.5	-22.2	94.5	-26.2	95.8
			-13.1	77.7	5.6	1.7	-17.2	78.9	-20.5	79.9	-23.6	80.8	-27.7	82.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.2	22.5	25.1	33.4	46.3	56.8	64.4	69.5	67.7	60.9	49.7	39.1	29.3
	DBStd	18.48	12.53	10.89	10.42	9.24	8.02	6.15	4.94	5.37	7.13	8.48	8.93	10.31
	HDD50	3385	853	699	525	177	26	0	0	0	3	114	344	644
	HDD65	6916	1317	1117	981	565	277	83	15	35	164	478	778	1106
	CDD50	2348	1	1	9	66	235	432	604	548	329	104	16	3
	CDD65	405	0	0	0	3	22	65	154	118	40	3	0	0
	CDH74	3756	0	0	3	81	287	642	1405	960	356	22	0	0
	CDH80	870	0	0	0	18	59	149	361	207	75	1	0	0
(14) Wind	WSAvg	4.9	5.3	5.9	6.2	6.5	4.9	4.0	3.3	3.1	3.7	4.9	5.3	5.5
(15) Precipitation	PrecAvg	47.9	3.5	3.0	3.7	3.7	4.2	4.4	4.4	4.1	4.1	4.3	3.9	3.9
	PrecMax	56.6	5.9	6.1	6.7	6.8	7.6	8.3	7.4	8.7	7.2	10.0	6.2	6.7
	PrecMin	36.5	1.7	1.8	1.7	1.2	1.4	1.7	1.9	1.2	2.0	0.4	2.5	1.6
	PrecStd	6.1	1.3	1.0	1.2	1.4	1.7	1.7	1.6	1.9	1.4	2.6	1.1	1.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	60.9	70.5	82.3	85.3	88.3	90.0	88.5	86.7	76.9	68.3	61.1
		MCWB	55.7	51.6	55.6	61.6	68.1	73.0	74.3	74.3	72.4	64.9	58.3	56.3
	2%	DB	51.0	52.6	60.5	75.0	81.2	84.2	86.6	84.6	81.7	72.6	63.4	54.2
		MCWB	45.6	44.3	50.8	57.1	65.7	70.2	73.0	71.5	69.2	63.1	55.2	49.8
	5%	DB	44.5	45.7	54.3	69.2	76.9	80.9	83.9	82.0	78.4	68.5	58.8	48.7
		MCWB	40.2	39.8	45.3	54.7	63.1	68.2	71.2	70.4	67.7	60.5	51.9	44.9
	10%	DB	39.3	40.6	49.4	63.5	72.8	77.5	81.5	79.6	74.2	64.1	54.4	44.4
		MCWB	35.6	36.0	42.5	51.5	61.0	65.6	70.0	68.8	65.2	57.2	48.9	40.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	54.4	57.1	65.5	71.1	74.5	77.1	76.2	74.0	68.3	60.6	56.7
		MCDB	60.1	60.3	65.4	76.1	82.2	84.7	86.7	84.9	83.1	74.1	66.2	60.8
	2%	WB	46.3	45.6	52.1	60.3	68.2	72.5	74.7	73.7	71.3	65.2	57.0	50.4
		MCDB	50.1	51.2	59.4	71.5	78.4	81.6	83.7	81.7	79.5	70.8	62.1	53.9
	5%	WB	40.6	40.2	46.8	56.7	65.5	70.4	73.0	72.1	69.0	61.6	53.2	45.4
		MCDB	44.2	45.6	52.5	67.4	74.2	78.3	81.5	79.7	75.8	67.6	57.7	48.7
	10%	WB	35.9	36.0	42.7	52.4	62.9	68.1	71.4	70.4	66.8	58.1	49.3	40.7
		MCDB	39.3	40.5	48.8	61.2	70.4	75.0	79.3	77.0	72.8	63.0	54.1	44.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	17.7	18.5	21.1	21.9	20.9	21.3	21.4	22.3	19.1	17.0	14.7
		MCDBR	21.7	22.7	26.2	30.9	28.4	25.4	23.9	23.4	25.6	25.1	23.4	19.9
	5% WB	MCWBR	18.5	17.5	17.1	16.4	15.3	13.4	12.1	12.2	14.4	15.8	16.9	17.7
		MCDWR	21.0	22.0	23.2	25.9	23.8	21.9	20.8	20.2	21.7	21.6	19.8	19.0
(40) Clear-Sky Solar Irradiance	taub		0.292	0.308	0.326	0.348	0.383	0.427	0.446	0.420	0.365	0.341	0.307	0.287
		taud	2.374	2.319	2.385	2.413	2.355	2.250	2.206	2.263	2.402	2.480	2.525	2.474
	Ebn at Noon		268	282	291	292	283	269	262	266	274	268	263	262
		Edh at Noon	25	31	33	35	38	43	44	41	33	27	22	21
(44) All-Sky Solar Radiation	RadAvg		505	783	1126	1431	1623	1756	1824	1632	1326	827	562	401
	RadStd		43	73	89	159	176	155	127	86	107	88	58	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (56 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+77

Nomenclature: See separate page